

Tharusha Sihan Fonseka

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EDUCATION

University of Moratuwa

Moratuwa, Sri Lanka

B.Sc. Engineering (Hons.) in Biomedical Engineering; CGPA: 3.94/4.00 🔗

Jun 2021 - Present

- Robotics, Autonomous Systems, Electronic Control Systems, Advanced Electronic Control Systems.
- Electronic Instrumentation, Embedded Systems, Medical Electronics and Instrumentation, Electronic Circuit Design.
- Digital Signal Processing, Bio Signal Processing, Medical Imaging, Data Structures and Algorithms.

Minor in Mathematics

May 2022 - Present

- Numerical Methods, Linear Algebra, Number Theory and Cryptography, Advanced Differential Equations, Linear Models and Multivariate Statistics, Calculus.

Ananda College

Colombo-10, Sri Lanka

GCE Advanced Level - Physical Science Stream; Z-Score 3.1996; Island Rank -1st

Jan 2017 - Aug 2019

- Combined Mathematics, Physics, Chemistry

Online courses

Robotics: Aerial Robotics (University of Pennsylvania, Coursera)

Oct 2023

ROS2 for Beginners (Udemy)

Apr 2025

EXPERIENCE

Research Affiliate - University of Sydney

Dec 2023 - May 2024

- Simulated the stretching of epidermal sensors with different body motions to analyze the stress and strain.
- Designed a wearable device to capture and transmit real-time resistance data from an epidermal stretch sensor.
- Setup the hardware for simultaneous data capturing from two radar devices in different frequency bands to create a synthetic band using machine learning.

Teaching Assistant - University of Moratuwa

2025

- Conducting electronic laboratory practicals of EN1094-Laboratory practice module.

AWARDS AND SCHOLARSHIPS

Dean's list

2022, 2023

Semesters - 1, 2, 3, 4, 5, 6, 7

Winners | Electronic Design Competition

Sep 2023

IEEE Sri Lanka section

1st Runners-up | SPARK Challenge 2023

Jun 2023

Innovation Competition for Sustainable Development - Electronic Club of University of Moratuwa

Dialog Merit Scholarship Award

2023

For outstanding performance in GCE A/L Examination.

Mahapola Higher Education (Merit) Scholarship

2021

For outstanding performance in GCE A/L Examination. Ranked #1 out of 19508 students

S.A. Wijethilake Memorial Challenge Cup | Ananda College

2020

Awarded for the Best Academic Performance of the College at the G.C.E. A/L Examination 2019

"Anada Pradeepa" Award | Ananda College

2020

Awarded to the College subject toppers based on the G.C.E. A/L Examination 2019

Silver medalist | Sri Lankan Physics Olympiad

2019

Awarded by Institute of Physics Sri Lanka

Silver medalist | Sri Lankan Astronomy and Astrophysics Olympiad

2018

Awarded by Institute of Physics Sri Lanka

PROJECTS

Humanoid Head For Robot Receptionist Capable of Natural Human Interaction	<i>Ongoing</i>
• Project lead. • The robot head is capable of bio mimicking facial expressions, active eye gazing, human re-identification, and natural conversation abilities. • I am working on hardware design, implementation, and ROS2-based controlling.	
Phoenix Cube Satellite Project	<i>Ongoing</i>
• Stage 1, launching a high-altitude balloon into the stratosphere. • I am leading a team of twenty undergraduates.	
Re-implementing research paper on Bio Signal Processing 	<i>2024</i>
• Re-implemented the paper "Classification of ECG heartbeats using nonlinear decomposition methods and support vector machine".	
Maze Solving Micromouse Robot(Robofest 2023, 2024) 	<i>2023, 2024</i>
• Fully Autonomous maze solving robot, which is tasked to reach the destination point in the shortest time possible. • I contributed to the sensor fusion, control system, and implementation of the maze-solving algorithm.	
Scorpion - Battle Robot (Exmo 2023) 	<i>Jul 2023</i>
• Designed a battle robot to demonstrate in EXMO 2023. • This includes a motorized hook and a linear roller, which were driven and controlled by an Arduino Nano. • I was involved in designing and implementing the mechanical design.	
Smart Switch for Automating Conventional Home Appliances 	<i>Jun 2023</i>
• Designed a smart switch to automate conventional home appliances, which can be easily integrated into an existing domestic installation. • Can be controlled remotely through internet or manually through capacitive touch buttons. • An individual project.	
Analog Implementation of a Five Band Audio Equalizer 	<i>Feb 2023</i>
• Five band audio equalizer using analog 4th order Sallen Key filters and a differential amplifier.	
IV Fluid Flow Regulator 	<i>Aug 2022</i>
• An automatic IV fluid flow regulator to aid in the treatment of dengue patients. • Flow rate and the period can be selected by the user. • Fluid is pumped using a peristaltic rotary pump.	

LEADERSHIP AND VOLUNTEERING

SEDS-Mora - University of Moratuwa <i>Project Manager</i>	<i>Apr 2024 - Present</i>
Old Anandian Engineers' Guild - Students' Chapter <i>Secretary</i> <i>Vice President</i> <i>Deputy Director- School Affairs</i>	<i>Mar 2024 - Present</i> <i>Mar 2023 - Mar 2024</i> <i>Jan 2022 - Mar 2023</i>
Brainstorm 2023 - IEEE EMBS, University of Moratuwa <i>Event Team Lead</i>	<i>Aug 2022 - May 2023</i>
Department of ENTC - University of Moratuwa <i>Field Representative</i>	<i>May 2022 - Feb 2023</i>
Roteract Club - University of Moratuwa <i>Event Co-chair - TourEye</i>	<i>Sept 2021</i>
Anandian Astronomical Association - Ananda College <i>Chief Organizer</i>	<i>Nov 2017 - Nov 2018</i>

SKILLS SUMMARY

Languages: English, Sinhala (Native Proficiency)

Programming Languages: Python, Matlab, C++

Software: PCB designing (Altium), CAD designing (Solidworks), Circuit simulation (Proteus, LtSpice)

Soft Skills: Leadership, Project Management, Team working

REFERENCES

Dr. Peshala Jayasekara

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Prof. Anusha Withana

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